

Port security operations

The backbone of shipping and business

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Operational Scale & Regulatory Framework: APSEZ & ISPS

APSEZ: The Infrastructure Giant

- **Scale:** 35,000 acres across 15 ports/terminals.
- **Reach:** Strategic presence on India's East & West coasts.

- **Market Share:** ≈27–28% of India's total port cargo.
- **Volume:** 450 MMT handled (12.5M TEUs container volume).

- **Annual Capacity:** 633 MMT cargo handling potential.
- **Status:** Largest private port developer in India.

01. Operational Footprint:

02. Performance (FY2025):

03. Capacity & Growth:

ISPS Code: The Global Mandate

01. Origin & Execution:

Under SOLAS, adopted in 2004 to strengthen maritime security against threats like terrorism and piracy

02. Scope of Application:

- Passenger ships, cargo ships over 500 gross tonnage, and port facilities serving international voyages
- Divided into Part A and B

03. The Core Elements

- Ship Security Plan
- Port Facility Security Plan
- Security Levels

Responsibilities, Roles & Guidelines:

1. The ISPS Officer Hierarchy

- **CSO (Company Security Officer):** Fleet-level oversight; manages Ship Security Plans (SSP) across the shipping company.
- **SSO (Ship Security Officer):** Vessel-level implementation; ensures crew compliance and on-board inspections.
- **PFSO (Port Facility Security Officer):** The anchor at Mundra Port; develops the security plan, conducts security drills, and acts as the liaison for ships.

2. National Pillar: CISF as the RSO

- **Designation:** The Central Industrial Security Force (CISF) is India's mandated Recognized Security Organisation (RSO) for ports.
- **The Unified Front:** CISF integrates with Port Authority, State Police, and the Indian Coast Guard for comprehensive multi-layered protection.

Guidelines

Pillar	Strategic Implementation
Legal Baseline:	• Governed by ISPS Code & ISO 28000. • Standard operations maintained at ISPS Level 1.
Multi-Level Cooperation:	• Active intelligence sharing with the Indian Coast Guard. • Mandatory threat reporting for all port stakeholders.
Resources & Human Element:	• Tech: Deployed AI, IVMS (In-Vehicle Monitoring), and ADAS. • Training: Strict enforcement of PPE and security vigilance.
Compliance:	• ISPS Certified until 2028 via continuous audits. • Proven adaptability to real-world threat alerts.

Port Facility Security Plan (PFSP) & Multi-Agency Coordination



The PFSP Framework

- **Origin:** Mandated by IMO under the ISPS Code (Post-9/11).
- **Definition:** The operational blueprint to protect ships, cargo, and infrastructure from sabotage and unauthorized access.
- **Core Objectives:**
 - Detect threats early & prevent unauthorized entry.
 - Ensure seamless inter-agency coordination.
 - Provide clear emergency incident procedures.
- **The Operational Shift:** Replaces 100% container inspection with Risk-Based Screening & random checks to maintain trade flow.

Application & Agencies

Execution: Controlled access systems, advanced CCTV surveillance, and strict ship-port interface monitoring.

The Government Security Matrix

Security at the port is a synchronized effort across multiple jurisdictions:

Agency	Operational Domain
Indian Coast Guard (ICG)	Maritime surveillance & offshore security.
CISF	Core infrastructure & port-entry security.
Customs (CBIC)	Cargo inspection & trade control.
Gujarat Marine Police	Coastal law enforcement.

Security Levels & Management Framework: The Dial of Vigilance

Security Level	Operational Status	Key Actions at Mundra Port
Level 1: Baseline	Normal (Steady)	PANS (96-hr notice), ID scans, Dolphin-19 boat patrols, live CCTV monitoring.
Level 2: Heightened	Intelligence-Led	Surprise patrols, random spot searches, restricted visitor access, Coast Guard liaison.
Level 3: Lockdown	Imminent Threat	Total access freeze, non-critical ops pause, QRT deployment, emergency-only entry.

The Operational Nerve Center

- **Integrated Security Control Room:** Central hub for all berths; 24/7 live feed & response coordination.
- **Declaration of Security (DoS):** Mandatory agreement between Ship & Port to synchronize security posture before berthing.

The "Resilience" Standard: ISO 28000

- **Beyond ISPS:** A total Security Management System (SMS).
- **Scope:** Focuses on risk assessment and continuous improvement across the entire supply chain, not just the port gate.

- **L1 (Perimeter):** Initial ID via Technology (Access Cards, Facial Recognition/FRS).
- **L2 (Intermediate):** Physical screening (HHMD, Mirror Trolleys) and bag checks.
- **L3 (Sensitive Zones):** Manual verification for restricted areas (Ro-Ro terminals/SEP areas).
- **Philosophy:** Overlapping controls ensure that a single failure (e.g., a lost card) does not lead to a total breach.

01. Defense in Depth (Layered Strategy)

- **Access Control:** Full-height turnstiles, flap barriers, and vehicle boom barriers.
- **Screening & Surveillance:** HHMDs, mirror trolleys, long-range binoculars, and speed guns.
- **Communication:** VHF Radios, Satellite phones, and Body Cameras for real-time accountability.

02. The Guard's Toolkit

- **QRT (Quick Response Team):** High-mobility units for rapid incident intervention.
- **RTMS (Road Traffic Management):** Ensuring clear corridors for emergency movement.
- **GMS/TMS: Guard Management Systems** ensuring patrol integrity and GPS-tracking.

03. Emergency & Accountability

Security Operations

Security Automation: The Digital Force Multiplier

1. The Operational Philosophy

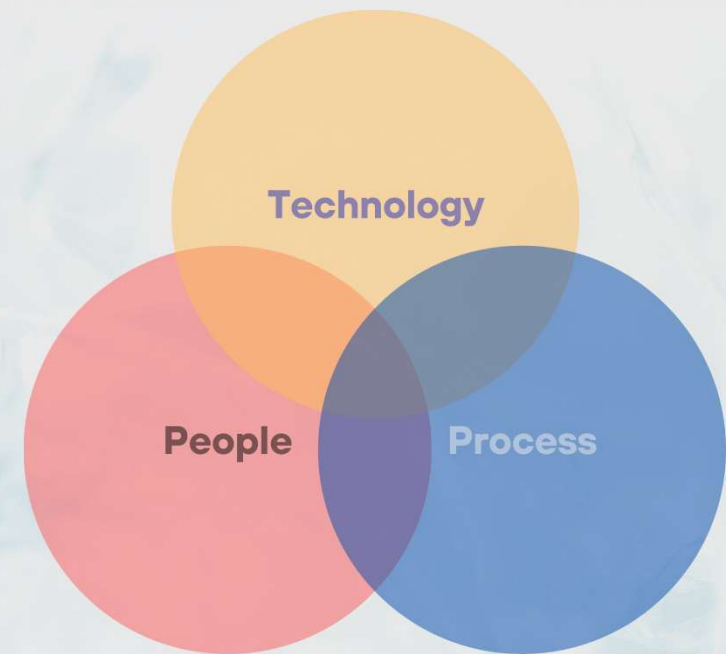
- **The Triad:** Process → People → Technology.
- **Efficiency:** Automating redundant human workflows via algorithmic processing.
- **The Conjugate:** Bridging the gap between physical security requirements and technical execution.

2. The Integrated Ecosystem (ISMS)

- **Genetec (The Core):** Unified platform for command, control, and endpoint management.
- **Kronos (The Identity):** HR-integrated logic defining user authority and access levels.
- **TechnoPurple (The Field):** Real-time GTMS (Guard Tool Management System) with GPS/Image verification.

3. Advanced AI & Analytics

- **AI Video Analytics:** Shifting from passive recording to active anomaly detection.
- **Integrated Hardware:** Autonomous response for Access Control, PIDs, and CCTV.



Emergency Response Protocols



01. Threat Classification & Core Framework

- **Threat Matrix:**
 - **On-Site:** Localized industrial/operational accidents.
 - **Off-Site:** Cyclones, earthquakes, tsunamis, bomb threats, or geopolitical conflict (war).
- **Incident Management Pillars:** Rapid coordination, effective communication, and mandatory logging/reporting.

02. Tactical Deployment & Resources

- **Command & Control:** Centralized Control Room for rapid reporting and emergency contacts.
- **Specialized Response Teams:**
 - **QRT (Quick Response Team):** Immediate, first-line incident response.
 - **Fire & Rescue:** Hazardous material and fire emergencies.
 - **Medical Team:** First aid and critical triage.
- **External Integration:** Seamless handover and coordination with Local Police Authorities.

03. Evacuation & Worker Safety

- Pre-defined Evacuation Procedures and safe Assembly Points.
- **STANDARD EMERGENCY SIREN CODES:**
 - **Declaration of Emergency:** 1 Minute Sounding → 5 Second Gap (Repeating)
 - **All Clear Signal:** Continuous 2 Minute Sounding

Comparative Analysis

- UVSS
- Hydrogen/ EV driven trucks
- Shover power
- Automated cranes and yard trucks
- AI-Driven Vessel Traffic Service

Conclusion

